# MAHARSHI OZA

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## PROFESSIONAL SUMMARY

Design Verification engineer with 4+ years of experience in digital IP verification and validation across ASIC/FPGA and automotive domains. Skilled in SystemVerilog, UVM, constrained-random verification, SVA, scoreboards, and functional/code coverage closure, with hands-on AXI4 and memory-subsystem verification. Strong foundation in RISC-V microarchitecture, out-of-order execution, and memory hierarchies.

## CORE COMPETENCIES

**Languages & Protocols:** SystemVerilog, Verilog HDL, C++, Python, Perl, Shell Scripting, AXI4, I2C, RISC-V ISA
**Design Verification:** UVM, Constrained-Random Verification (CRV), SVA, Functional & Code Coverage, Coverage Closure, Regression Testing, Debug & Root-Cause Analysis, Scoreboards, TLM, SoC/IP Verification
**RTL & ASIC:** RTL Design, Logic Synthesis, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Design for Testability (DFT)
**Tools:** VCS, Xcelium, QuestaSim, ModelSim, Design Compiler, Vivado, Git, Linux
**Computer Architecture:** RISC-V, Pipelining, Superscalar & Out-of-Order Execution, Register Renaming, Branch Prediction, Memory Hierarchy
**GPU & AI Hardware:** GPU/SIMT Architecture, CUDA, GPU Memory Hierarchy, Warp Execution, Parallel Computing, Matrix Multiplication, Convolution, Parallel Reduction, AI/LLM Accelerators

## PROFESSIONAL EXPERIENCE

**RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India** **Aug 2023 – Aug 2024**

- Implemented an out-of-order scoreboard for an AXI4 memory subsystem in SystemVerilog/UVM, tracking outstanding transactions by ID and checking read/write data, responses, and per-ID ordering rules.
- Developed functional coverage for AXI4 burst types, lengths, sizes, IDs, out-of-order completion, and handshake stalls, achieving over 90% functional coverage; maintained UVM environments for Asynchronous FIFO and FPU blocks.
- Verified corner cases including reset/recovery, FIFO full/empty and overflow/underflow, data integrity, and AXI handshake correctness; debugged regressions using Questa, Xcelium, VCS, and Vivado to isolate RTL and testbench issues.
- Contributed to the RTL design of a multi-port packet router (FSM control, internal FIFOs, synchronizers) and aligned verification scenarios with ongoing RTL and specification updates.

**Functions Engineer – Vision & Perception | Magna Electronics, India** **Dec 2021 – Aug 2023**

- Developed validation code for ADAS 2+ lane-marker detection, checking lane width, inter-lane distance, and related geometry against ground truth, and built KPI evaluation utilities for safety-critical perception features.
- Implemented code, branch, and line coverage for 2 perception modules and the KPI framework to expose validation gaps; root-caused edge-case failures with GDB and applied MISRA C practices in development and review.

## PROJECTS

**UVM Verification of a Set-Associative Write-Back Cache Controller | SystemVerilog, UVM, SVA, Python**

- Built a UVM 1.2 environment for a parameterized write-back/write-allocate cache controller with 3 agents (CPU, flush, reactive memory responder via TLM analysis FIFO), a reference-model scoreboard, and SVA protocol checks split into testbench-side and DUT-side assertions.
- Developed 7 directed and constrained-random tests using factory config overrides (backpressure, flush/CPU races, memory error injection); built a Python regression flow merging per-run coverage and reporting holes, with 0 data mismatches across 1,000+ accesses under injected errors.

**Functional Verification of LC-3 Processor | UVM/UVMF**

- Architected a reusable UVM/UVMF environment covering LC-3 pipeline stages and instruction/data memory; developed sequences, tests, assertions, regression, and coverage models for control-flow, datapath, and corner cases, achieving 100% functional coverage closure.
- Built independent scoreboard-based checking to validate instruction execution against expected pipeline behavior, enabling systematic bug isolation across fetch, decode, and execute stages.

**UVM-Based Verification of AXI4 Slave | SystemVerilog**

- Built a UVM environment for a 32-bit AXI4 slave supporting FIXED/INCR bursts; implemented agents, scoreboard, SVA protocol checks, and functional coverage for burst types, sizes, and boundary conditions, including constrained-random backpressure scenarios.
- Validated protocol compliance under stall and backpressure conditions using SVA-based checks, catching timing violations.

**Functional Verification of I2C Multi-Bus Controller | SystemVerilog**

- Constructed a layered ncsu_pkg-based verification environment with generator, predictor, scoreboard, assertions, and coverage; exercised protocol behavior and corner cases using functional coverage and SVA, achieving 93% coverage.

**Stride Value Predictor for RISC-V OOO Pipeline Simulator | C++**

- Developed a cycle-accurate RISC-V out-of-order simulator and integrated a Stride Value Predictor with VPQ/checkpoint recovery; added a Global Stride History Table with LRU replacement, improving harmonic-mean IPC from 1.91 to 2.00 across 15 benchmarks within a 27 KB storage budget.

## EDUCATION

**M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4** **Aug 2024 – May 2026**
**Relevant Coursework:** Advanced Verification using UVM, ASIC Verification, ASIC and FPGA Design with Verilog, Advanced Microprocessor Architecture, Hardware–Software Codesign, Physical Design, Microarchitecture Reverse Engineering

**B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India** **Aug 2018 – May 2022**